CSIR: Council of Scientific & Industrial Research

Project : Smart Anomaly Detection In Radiology Scans Using YOLO-Based Detection Models

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Abstract :

In today’s fast-paced healthcare environment, early and accurate detection of abnormalities in radiology scans is crucial. This project introduces an intelligent system that uses deep learning to automatically detect unusual patterns or anomalies in medical images like X-rays, CT scans, and MRIs. By using YOLO (You Only Look Once) . The system can quickly and precisely highlight areas of concern in the scans. This not only speeds up the diagnostic process but also helps doctors by reducing the chances of missing critical findings. The goal is to create a supportive tool that brings both speed and accuracy to medical imaging analysis.

Introduction :

- Project lies in the field of **Radiology + Artificial Intelligence (AI)**
- Focused on detecting **anomalies in medical images** (X-rays, CTs, MRIs)

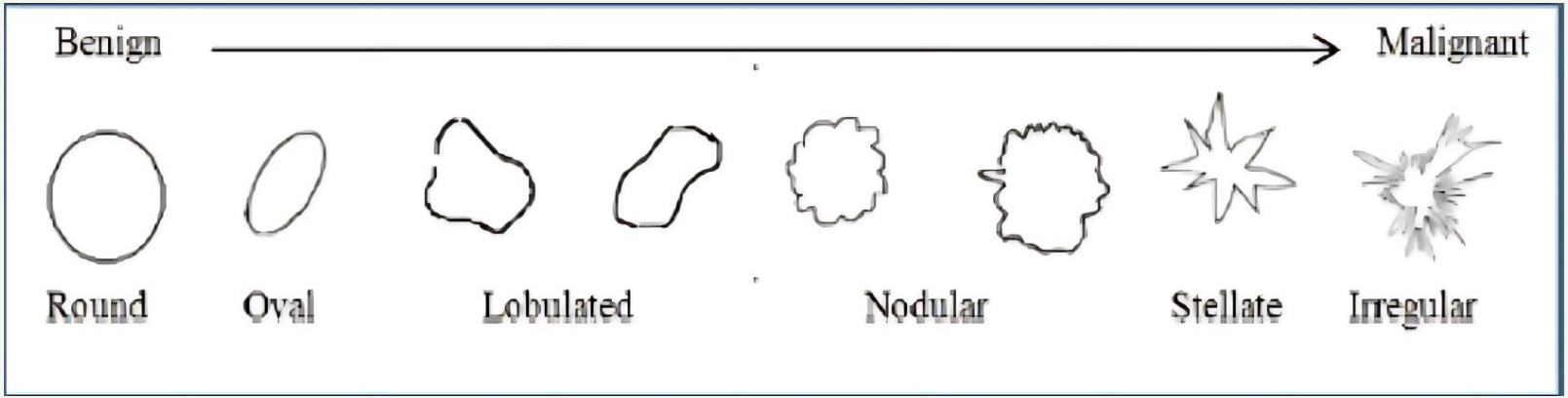
- Uses **Computer Vision** techniques for smart medical analysis
- Enhances diagnostic speed, **accuracy**, and overall patient care

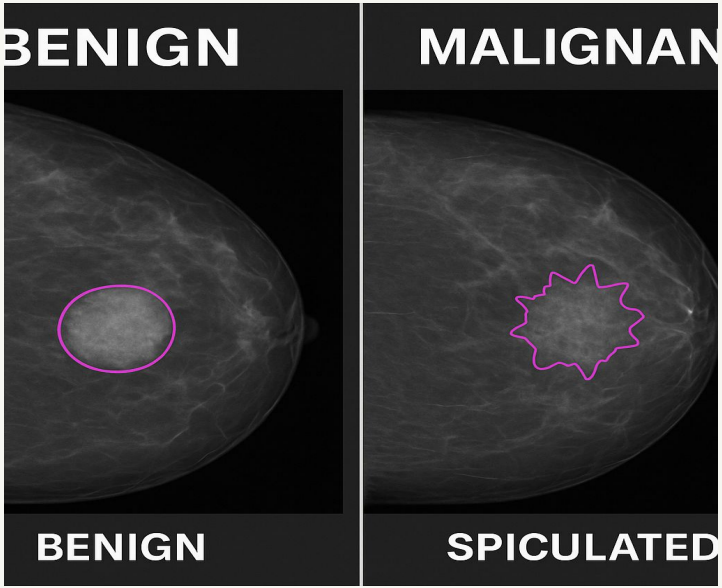
Problem Statement :

- Manual detection of anomalies in radiology scans is time-consuming and prone to human error, which can delay diagnosis and affect treatment quality.

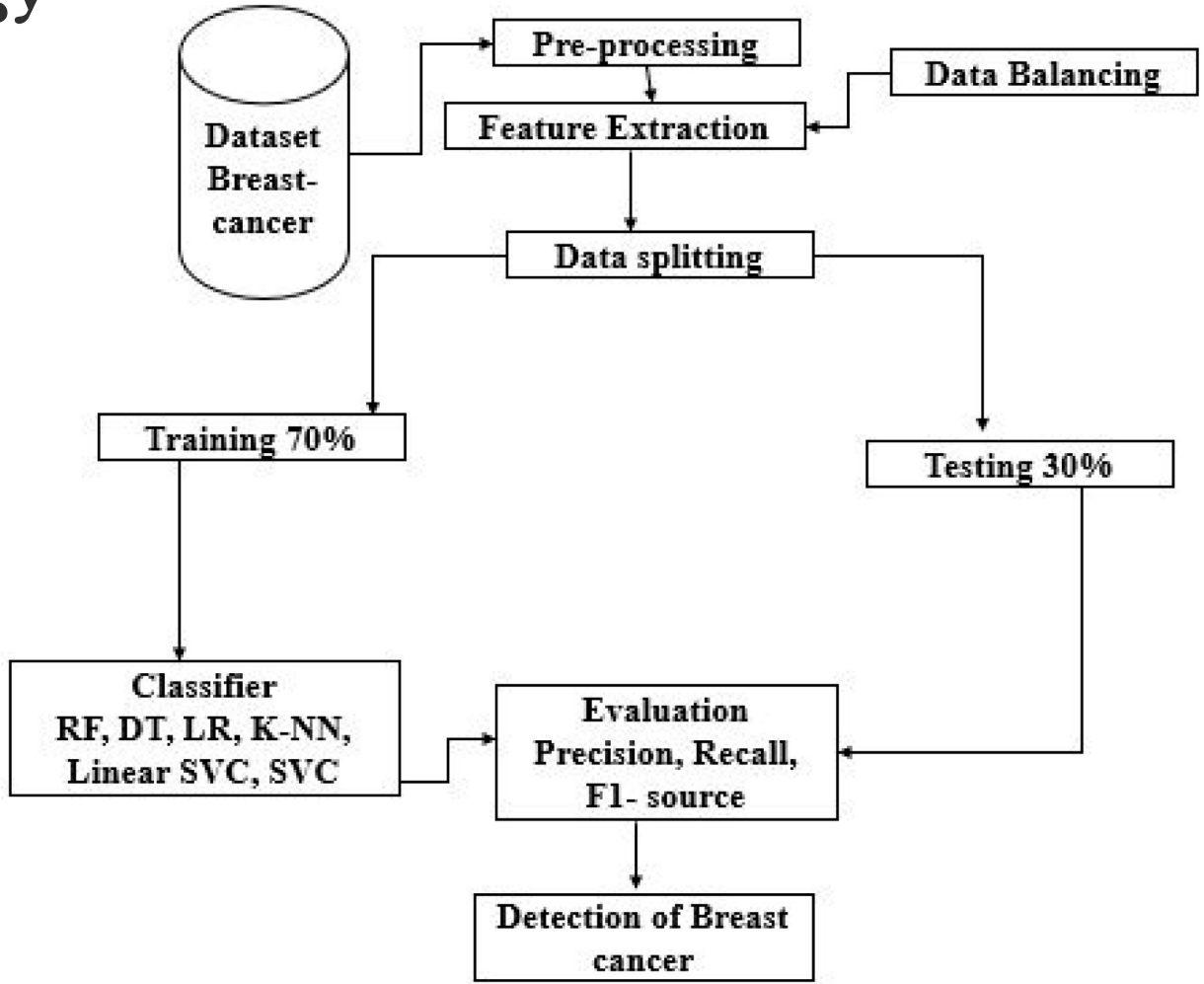
- Early and accurate diagnosis is critical in medical cases like tumors or internal injuries. Any delay or oversight can lead to serious health risks.

- By using a YOLO-based deep learning model, the system can automatically detect and localize anomalies in radiology scans with high speed and accuracy, assisting doctors in making faster decisions.


Benign
Malignant
Round
Oval
Lobulated
Nodular
Stellate
Irregular


BENIGN
MALIGNAN
BENIGN
SPICULATED

Methodology



Design & Implementation details

- **Environment Setup:** Installed **YOLOv8 (Ultralytics)** in **Google Colab** with GPU support.

- **Dataset Preparation:** Collected and annotated medical datasets using **Roboflow**.

- **Model Training:**
  - Input size: **640×640**
  - Epochs: **50**
  - Batch size: **16**

- **Evaluation:** Model performance measured using **mAP**, **Precision**, **Recall**, and **IoU** metrics.

- **System Integration:** Integrated the trained model with Google colab for image uploads and real-time anomaly detection.

Train code

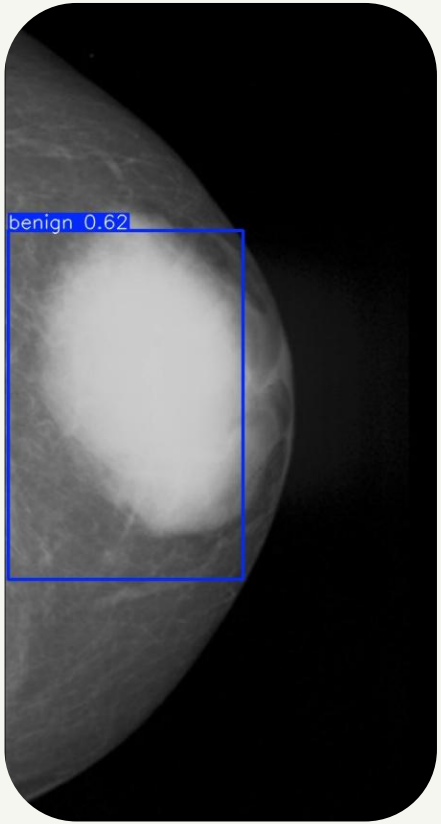
```python
from ultralytics import YOLO

model = YOLO('yolov8n.pt')
model.train(
    data='/content/drive/My Drive/data/config.yaml',
    epochs=50,
    name='cancer_cell',
    project='/content/drive/My Drive/trained'
)a
```

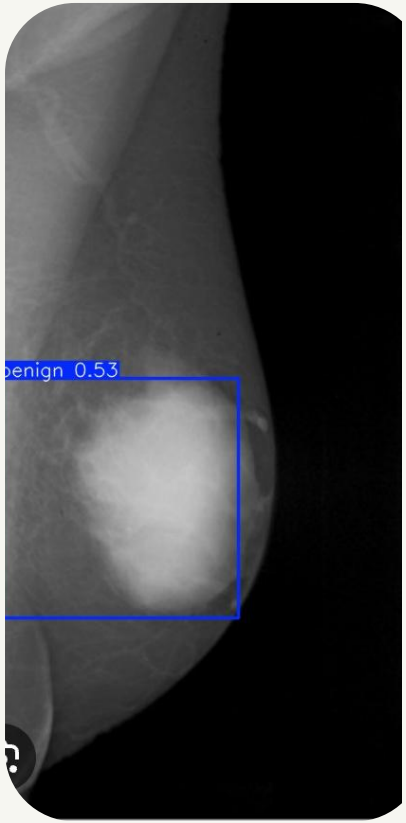
Model Train
Compile live

```
Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
43/50      2.96G      1.567      1.446       1.55         10        640: 100%|          | 99/99 [02:38<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
44/50      2.96G      1.563      1.424      1.554          9        640: 100%|          | 99/99 [02:37<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
45/50      2.99G      1.539      1.412      1.543          8        640: 100%|          | 99/99 [02:37<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
46/50      3.01G      1.538      1.396      1.537         11        640: 100%|          | 99/99 [02:36<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
47/50      3.03G      1.517      1.386      1.516          9        640: 100%|          | 99/99 [02:39<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
48/50      3.04G      1.512      1.381      1.522         12        640: 100%|          | 99/99 [02:36<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
49/50      3.06G      1.502      1.363      1.507          9        640: 100%|          | 99/99 [02:14<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13

Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
50/50      3.08G      1.492      1.345      1.507          8        640: 100%|          | 99/99 [02:33<
           Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%|          | 13
```

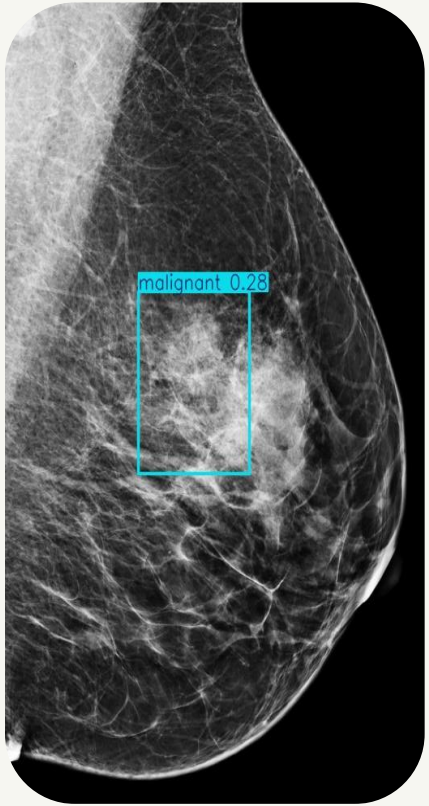
Result

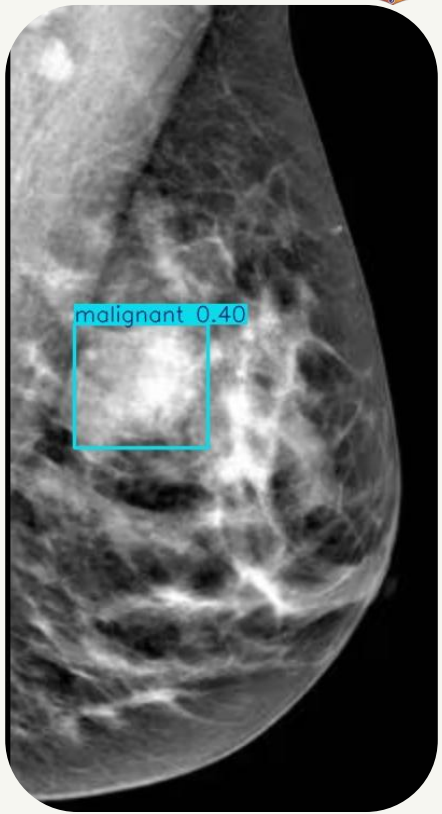
BENIGN





Malignant





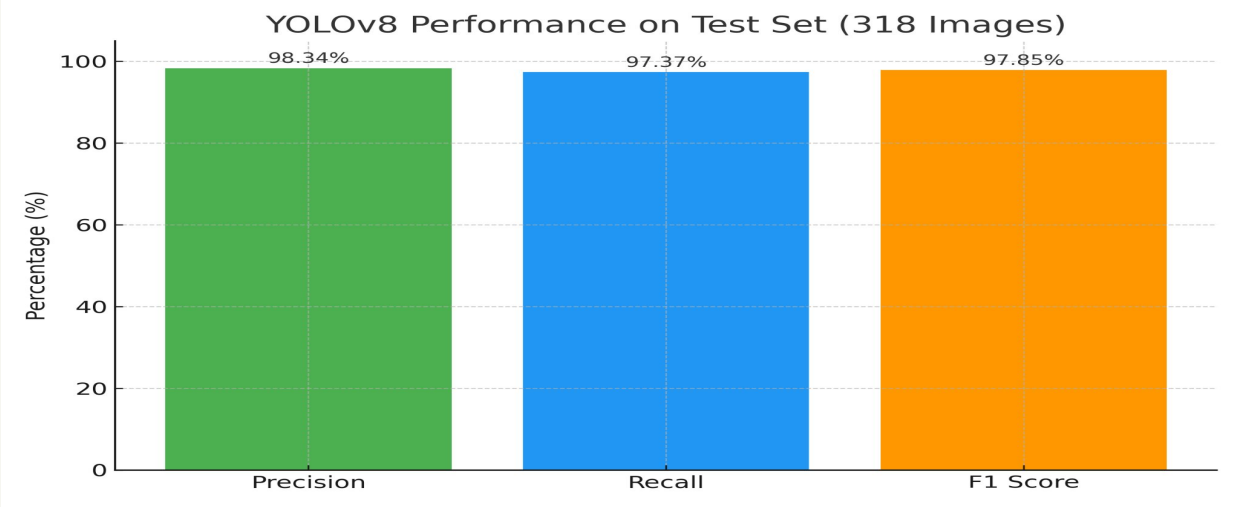
Evaluation Metrics

Model	Precision (%)	Recall (%)	F1 Score (%)	Inference Time (s for 318 images)	Accuracy
YOLOv8	98.34	97.37	97.85	11.98	95.6%

- Best balance of **high accuracy (95.6%)**, **real-time speed**.

Metric and Graph

Metric	YOLOv8 Value	Test Set Size	Inference Time (s)
Precision	98.34%	318 images	11.98
Recall	97.37%	318 images	11.98
F1 Score	97.85%	318 images	11.98



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THANK YOU

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